compute_dissimilarity_matrix.R

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`compute_dissimilarity_matrix` *Compute Dissimilarity Matrix from Trait Data*

Description

Computes a pairwise dissimilarity matrix from a wide trait table using ‘cluster::daisy()‘. Before computation, any ‘Inf‘ / ‘-Inf‘ values in numeric trait columns are replaced with ‘NA‘ so that the normalisation denominator stays finite. After computation, any remaining ‘NaN‘ or non-finite values in the distance matrix are replaced with ‘1.0‘ (the maximum dissimilarity, i.e. fully dissimilar). This function isolates the distance-computation step from ‘cluster_functional_types()‘ so that the distance matrix can be stored as an independent, inspectable pipeline target.

Usage

```
compute_dissimilarity_matrix(data, taxon_col = "taxon_name", metric = "gower")
```

Arguments

`data`	A data frame with one row per taxon. Must contain a column identified by ‘taxon_col‘ and at least one additional trait column. ‘NA‘ values in trait columns are handled natively by ‘cluster::daisy()‘.
`taxon_col`	A single character string naming the column that holds taxon names. Default: ‘"taxon_name"‘. This column is excluded before the distance computation.
`metric`	A single character string passed to ‘cluster::daisy()‘ as the ‘metric‘ argument. Default: ‘"gower"‘. Other valid values are ‘"euclidean"‘ and ‘"manhattan"‘ (see ‘?cluster::daisy‘).

Details

Steps performed:

1. Select all columns except ‘taxon_col‘ as trait columns.
2. Replace any ‘Inf‘ / ‘-Inf‘ values in numeric trait columns with ‘NA‘ via ‘dplyr::if_else()‘.
3. Compute ‘cluster::daisy(metric = metric)‘.
4. If any value in the resulting distance vector is non-finite (e.g. ‘NaN‘ arising when two taxa share no non-‘NA‘ trait), replace it with ‘1.0‘ and reconvert via ‘stats::as.dist()‘.

Value

An object of class ‘"dist"‘ (as returned by ‘stats::as.dist()‘) with one entry per pair of taxa. All values are in [0, 1] when ‘metric = "gower"‘ (Gower default).

See Also

[fit_hclust()], [cluster_functional_types()]

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